

On eigenvalues of a $\mathcal{PT}$ -symmetric operator in a thin layer

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Abstract. We consider an elliptic operator with variable coefficients in a thin three-dimensional layer with $\mathcal{PT}$ -symmetric boundary conditions. We study the effect of the appearance of isolated eigenvalues at the edges of the gaps in the essential spectrum. We obtain sufficient conditions that guarantee that such eigenvalues either exist or are absent near a given edge of a gap. In the case of existence, the first terms in the asymptotic expansion of these emerging eigenvalues are calculated.

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§ 1. Introduction

The theory of $\mathcal{PT}$ -symmetric operators has been developing fairly intensively in the last two decades (see, for example, [1]–[11] and the survey of the literature in these papers). These studies motivate the study of a model of $\mathcal{PT}$ -symmetric operators on segments, in layers and strips, where the property of being $\mathcal{PT}$ -symmetric is brought about primarily by the boundary conditions (see [12]–[16]). These papers considered Robin boundary conditions with a purely imaginary coefficient. The case when the width of the strip or the layer is small was studied in [14]–[16]. The importance of this case is explained by the lasting interest shown in thin domains (see, for example, [17]–[22] and the bibliography in these papers). Problems in thin strips, layers, and cylinders often arise in modelling nanostructures too (see [20], [22]). Therefore, studying $\mathcal{PT}$ -symmetric operators in thin domains seems to be of great interest.

In [14] the Laplace operator was considered in a thin multi-dimensional layer with $\mathcal{PT}$ -symmetric boundary conditions

$$\left(\frac{\partial}{\partial x_n} + i\alpha \right) u = 0, \quad (1.1)$$

where x_n is the transverse variable in the layer and α is a certain given real bounded sufficiently smooth function. A similar model was considered in [15], but

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the Laplace operator was replaced by a more general scalar elliptic differential operator of the second order with variable coefficients. Certain conditions of parity type were imposed on these coefficients, which, together with the boundary conditions, ensured that the operator was $\mathcal{PT}$ -symmetric. The main results in [14] and [15] are a description of the effective operator, a proof that the resolvent of the original operator converges to the resolvent of the effective operator in a certain operator norm, and estimates of the rate of convergence of the resolvents. The convergence of the spectrum of the original operator to the spectrum of the effective operator was proved on the basis of resolvent convergence. Furthermore, in [15] complete asymptotic expansions were constructed for the eigenvalues of the original operator converging to isolated limit eigenvalues with respect to a small parameter, which is the width of the layer. Similar expansions were also obtained for the associated eigenfunctions.

Besides the eigenvalues that remain isolated in the limit, the original operator in a thin domain may have eigenvalues converging to the edges of the essential spectrum. This effect is well known for small regular perturbations of selfadjoint operators (see, for example, [23]–[28]). However, research into this effect for thin domains has only recently begun. In thin strips and three-dimensional layers with variable width, eigenvalues in gaps of the continuous spectrum were discussed for the Laplace operator and for the operator of elasticity theory in [29]–[31]. The question of the existence of such eigenvalues was studied and their asymptotic behaviour was described. For $\mathcal{PT}$ -symmetric operators, this effect was studied in the recent paper by one of the authors [16], where the Schrödinger operator with decreasing potential in a thin strip was considered. Boundary conditions were chosen in the form (1.1), where α was assumed to be a constant. It was established that under certain sufficient conditions, in a neighbourhood of the lower threshold of the essential spectrum an isolated eigenvalue may appear, which converges to this boundary as the width of the strip decreases. It was shown that such an eigenvalue is unique, real and simple. The first terms of the asymptotic expansion were obtained for this eigenvalue with respect to a small parameter—the width of the strip. Sufficient conditions for such an eigenvalue to exist were also derived.

In this paper we continue the studies initiated in [16]. We consider an elliptic operator with variable coefficients in a thin three-dimensional layer. The coefficients of the higher derivatives depend only on the longitudinal variables; the potential may depend on the transversal variable in addition. The boundary conditions again have the form (1.1), but this time α is the sum of a constant and a decreasing function. We show that the essential spectrum of such an operator has a banded structure and may contain gaps. The main result of our paper is a description of isolated eigenvalues in a neighbourhood of the edges of such gaps. We present sufficient conditions that guarantee the existence or absence of such eigenvalues near a given edge of a gap. In the case of existence, we prove that the emerging eigenvalue is unique, real and simple. For a given eigenvalue we obtain the first terms of the asymptotic expansion with respect to the width of the layer. It turns out that, in contrast to the polynomial asymptotics in [16], here the asymptotics are exponential. Furthermore, the more complicated form of the operator and the increase in the dimension of the domain require a serious elaboration of the technique of [16]. In particular, this is related to the form of the effective operator.

The latter now is a periodic two-dimensional operator with variable coefficients to which a potential is added that is decreasing as a power function. In [16] the effective operator was the one-dimensional Schrödinger operator with exponentially decreasing potential.

The paper is organized as follows. In § 2 we pose the problem and state the main results. In § 3 we describe the essential spectrum. Section 4 is devoted to studying the effect of the emergence of isolated eigenvalues. We put the proofs of all the auxiliary lemmas in a separate section (§ 5), because they are quite lengthy.

§ 2. A statement of the problem and the results

Let $x' = (x_1, x_2)$, $x = (x', x_3)$ be Cartesian coordinates in $\mathbb{R}^2$ and $\mathbb{R}^3$, respectively, let ε be a small positive parameter, and $\Omega_\varepsilon := \{x: 0 < x_3 < \varepsilon\}$ a thin infinite layer. In the domain Ω_ε we consider the elliptic operator

$$\mathcal{H}_\varepsilon := - \sum_{i,j=1}^2 \frac{\partial}{\partial x_i} A_{ij} \frac{\partial}{\partial x_j} - \frac{\partial^2}{\partial x_3^2} + A_0 + V_\varepsilon$$

with the $\mathcal{PT}$ -symmetric boundary conditions

$$\left(\frac{\partial}{\partial x_2} + i\alpha \right) u = 0 \quad \text{on } \partial\Omega_\varepsilon. \quad (2.1)$$

Here, $A_{ij} = A_{ij}(x')$, $A_0 = A_0(x')$, $A_{ij} \in C^1(\mathbb{R}^2)$ and $A_0 \in C(\mathbb{R}^2)$ are given functions, which are real, and periodic with respect to the integer lattice $\mathbb{Z}^2$. The functions A_{ij} are assumed to satisfy the ellipticity condition

$$\sum_{i,j=1}^2 A_{ij}(x') \xi_i \xi_j \geq c_0 (\xi_1^2 + \xi_2^2), \quad x' \in \mathbb{R}^2, \quad (\xi_1, \xi_2) \in \mathbb{R}^2,$$

where c_0 is some constant independent of x' and (ξ_1, ξ_2) .

The potential V_ε is introduced by the equation

$$V_\varepsilon(x) = V\left(x', \frac{x_3}{\varepsilon}\right),$$

where $V(x', \xi)$ is a complex-valued function continuous in $\mathbb{R}^2 \times [0, 1]$ and satisfying the uniform estimate

$$|V(x', \xi)| \leq C \chi_1(x') \chi_2(x') \quad \text{in } \mathbb{R}^2 \times [0, 1], \quad (2.2)$$

where C is some constant independent of x and ξ , and $\chi_1 = \chi_1(x')$, $\chi_2 = \chi_2(x')$ are infinitely differentiable non-negative functions such that

$$(1 + |x'|) \chi_1 \in L_2(\mathbb{R}^2) \cap L_\infty(\mathbb{R}^2), \quad (2.3)$$

$$(1 + |x'|^2) \chi_2 \in L_2(\mathbb{R}^2), \quad \lim_{x' \rightarrow \infty} \chi_2 = \lim_{x' \rightarrow \infty} \frac{\partial \chi_2}{\partial x_i} = \lim_{x' \rightarrow \infty} \frac{\partial^2 \chi_2}{\partial x_i \partial x_j} = 0. \quad (2.4)$$

The additional restriction

$$V(x_1, 1 - \xi) = \overline{V(x_1, \xi)} \quad (2.5)$$

is imposed on the function V .

In boundary condition (2.1), i denotes the imaginary unit, and α a real function

$$\alpha(x') = \alpha_0 + \alpha_1(x'),$$

where $\alpha_0 \in \mathbb{R}$ is a given constant, while $\alpha_1 \in C^2(\mathbb{R}^2)$ is a given real function satisfying

$$|\alpha_1(x')| + \sum_{i=1}^2 \left| \frac{\partial \alpha_1}{\partial x_i}(x') \right| + \sum_{i,j=1}^2 \left| \frac{\partial^2 \alpha_1}{\partial x_i \partial x_j}(x') \right| \leq C \chi_1(x') \chi_2(x'), \quad (2.6)$$

where the constant C is independent of x' .

The operator $\mathcal{H}_\varepsilon$ is treated as an unbounded operator in $L_2(\Omega_\varepsilon)$ on the domain of definition

$$\mathcal{D}(\mathcal{H}_\varepsilon) := \{u \in W_2^2(\Omega_\varepsilon) : \text{condition (2.1) holds}\}.$$

We emphasize that such an operator is neither selfadjoint nor symmetric. At the same time, in accordance with Theorem 2.1 in [15], the operator $\mathcal{H}_\varepsilon$ is m -sectorial, $\mathcal{T}$ -selfadjoint, and $\mathcal{P}$ -pseudo-Hermitian, namely,

$$\mathcal{H}_\varepsilon^* = \mathcal{T} \mathcal{H}_\varepsilon \mathcal{T}, \quad \mathcal{H}_\varepsilon^* = \mathcal{P} \mathcal{H}_\varepsilon \mathcal{P},$$

where $\mathcal{T}$ is the operator of complex conjugation: $(\mathcal{T}u)(x) = \overline{u(x)}$, and $\mathcal{P}$ is the operator of reflection: $(\mathcal{P}u)(x) = u(x_1, \varepsilon - x_2)$. The standard $\mathcal{PT}$ -symmetry condition also holds:

$$\mathcal{P} \mathcal{T} \mathcal{H}_\varepsilon = \mathcal{H}_\varepsilon \mathcal{P} \mathcal{T}.$$

Our goal is to study the behaviour of isolated eigenvalues of the operator $\mathcal{H}_\varepsilon$ near the edges of the gaps in the essential spectrum for small ε .

The essential spectrum of an operator is defined in terms of characteristic sequences (see [32], Ch. I, § 1.1) and is denoted by $\sigma_{\text{ess}}(\cdot)$. To describe the essential spectrum of the operator $\mathcal{H}_0$ we introduce the operator

$$\mathcal{H}_0 := - \sum_{i,j=1}^2 \frac{\partial}{\partial x_i} A_{ij} \frac{\partial}{\partial x_j} + A_0$$

in $L_2(\mathbb{R}^2)$ on the domain of definition $W_2^2(\mathbb{R}^2)$. This operator is periodic and selfadjoint. For an arbitrary set $X \subset \mathbb{R}$ and a constant $a \in \mathbb{R}$, let $X + a$ denote the shift of X by a : $X + a := \{x \in \mathbb{R} : x - a \in X\}$.

Our first result describes the essential spectrum of the operator $\mathcal{H}_\varepsilon$.

Theorem 1. *The essential spectrum of the operator $\mathcal{H}_\varepsilon$ has the form*

$$\sigma_{\text{ess}}(\mathcal{H}_\varepsilon) = \sigma_{\text{ess}}(\mathcal{H}_0) + \alpha_0^2 + \bigcup_{j=1}^{\infty} \left(\sigma_{\text{ess}}(\mathcal{H}_0) + \frac{\pi^2 j^2}{\varepsilon^2} \right).$$

Note that if the Bethe-Sommerfeld conjecture holds for the operator $\mathcal{H}_0$ and its spectrum has only finitely many gaps, then for sufficiently small ε the formula for the spectrum of the operator $\mathcal{H}_\varepsilon$ in Theorem 1 becomes significantly simpler:

$$\sigma_{\text{ess}}(\mathcal{H}_\varepsilon) = \sigma_{\text{ess}}(\mathcal{H}_0) + \alpha_0^2.$$

The essential spectrum of the operator $\mathcal{H}_0$ is described using Floquet-Bloch theory. Namely, let $\square := (0, 1)^2$ be a periodicity cell of the lattice $\mathbb{Z}^2$, and let $\square^* := (0, 2\pi)^2$ be a Brillouin zone. We define the family of operators

$$\mathcal{H}_0(\tau) := \sum_{i,j=1}^2 \left(i \frac{\partial}{\partial x_i} + \tau_i \right) A_{ij} \left(i \frac{\partial}{\partial x_j} + \tau_j \right) + A_0, \quad \tau = (\tau_1, \tau_2) \in \square^*,$$

in $L_2(\square)$ with periodic boundary conditions. The domain of definition of such an operator is the Sobolev space $W_{2,\text{per}}^2(\square)$ consisting of functions in $W_2^2(\square)$ satisfying periodic boundary conditions.

The operator $\mathcal{H}_0(\tau)$ has compact resolvent, and its spectrum consists of isolated eigenvalues of finite multiplicity, which we arrange in increasing order, taking account of multiplicities:

$$E_1(\tau) \leq E_2(\tau) \leq E_3(\tau) \leq \dots$$

The essential spectrum of the operator $\mathcal{H}_0$ is determined by the equation

$$\sigma_{\text{ess}}(\mathcal{H}_0) = \bigcup_{n=1}^{\infty} \{E_n(\tau) : \tau \in \overline{\square^*}\}$$

(see [33], Ch. 4, §4.5, Theorem 4.5.1). Suppose that this spectrum has a gap (β_-, β_+) ; we do not exclude the situation where $\beta_- = -\infty$, which corresponds to a semi-infinite gap. Then by Theorem 1 for sufficiently small ε the essential spectrum of the operator $\mathcal{H}_\varepsilon$ contains the gap $(\beta_- + \alpha_0^2, \beta_+ + \alpha_0^2)$. Our next goal is to describe the isolated eigenvalues of the operator $\mathcal{H}_\varepsilon$ in a neighbourhood of the edges of this gap.

We begin with the right-hand edge of the gap, for which we make the following assumption.

- A1) The edge of the gap β_+ is a minimum of only one band function E_m . This minimum is nondegenerate and is attained at an interior point τ_0 of the Brillouin zone $\square^*$. The eigenvalue $E_m(\tau_0)$ is a simple eigenvalue of the operator $\mathcal{H}_0(\tau_0)$. The corresponding eigenfunction ϕ_0 does not vanish identically on $\partial\square$.

Because the minimum of the function E_m is nondegenerate at the point τ_0 , as $\tau \rightarrow \tau_0$ we have the relation

$$E_m(\tau) = \beta_+ + E_m''(\tau - \tau_0) + O(|\tau - \tau_0|^3), \quad (2.7)$$

where E_m'' is a positive-definite quadratic form. We set

$$d_+ := \frac{1}{\sqrt{\det E_m''}}, \quad V_0(x') := 2\alpha_0\alpha_1(x') + \alpha_1^2(x') + \int_0^1 V(x', \xi) d\xi,$$

where $\det E_m''$ is the determinant of the matrix of the quadratic form E_m'' .

We assume that one of the following assumptions holds.

A2) The equation

$$\left(- \sum_{i,j=1}^2 \frac{\partial}{\partial x_i} A_{ij} \frac{\partial}{\partial x_j} + A_0 + V_0 - \beta_+ \right) \psi_0 = 0 \quad \text{in } \mathbb{R}^2 \quad (2.8)$$

has a unique bounded nontrivial solution. This solution has the following asymptotics at infinity:

$$\psi_0(x') = e^{i\tau_0 \cdot x'} \phi_0(x') + o(1), \quad x' \rightarrow \infty, \quad (2.9)$$

where “ $\cdot$ ” is the inner product in $\mathbb{R}^2$ and the function ϕ_0 is chosen to be normalized in $L_2(\square)$.

A3) Equation (2.8) does not have nontrivial bounded solutions.

When assumption A2) holds, we set

$$\begin{aligned} V_1(x') &:= 2\alpha_0 \int_0^1 (2\xi - 1) \operatorname{Im} V(x', \xi) dx', \\ V_2(x') &:= -\alpha_0^2 \int_0^1 \left(2\xi^2 - 2\xi + \frac{1}{3} \right) \operatorname{Re} V(x', \xi) d\xi, \end{aligned} \quad (2.10)$$

$$V_3(x') := \sum_{i,j=1}^2 A_{ij}(x') \frac{\partial \alpha_1}{\partial x_i}(x') \frac{\partial \alpha_1}{\partial x_j}(x'), \quad (2.11)$$

$$\mu_1 := -\frac{d_+}{4\pi^2} \int_{\mathbb{R}^2} V_1 |\psi_0|^2 dx', \quad (2.12)$$

$$\mu_2 := -\frac{d_+}{4\pi^2} \int_{\mathbb{R}^2} \left(V_2 + \frac{V_3}{3} \right) |\psi_0|^2 dx' + \frac{d_+}{4\pi^2} \int_{\mathbb{R}^2} \bar{\psi}_0 V_1 U_0 dx', \quad (2.13)$$

where U_0 is a solution of the equation

$$\left(- \sum_{i,j=1}^2 \frac{\partial}{\partial x_i} A_{ij} \frac{\partial}{\partial x_j} + A_0 + V_0 - \beta_+ \right) U_0 = V_1 \psi_0 \quad \text{in } \mathbb{R}^2$$

with the asymptotics

$$\begin{aligned} U_0(x') &= -2\pi\mu_1 e^{i\tau_0 \cdot x'} \phi_0(x') (-\ln|x'| + c_1) + O(|x'|^{-1}), \\ c_1 &:= \int_0^1 \frac{J_0(t) - 1}{t} dt + \int_1^{+\infty} \frac{J_0(t)}{t} dt, \end{aligned}$$

where J_0 is the Bessel function. Below, in Lemma 1 we will show that the function U_0 is well defined and unique.

Our main result, which follows, describes the isolated eigenvalues of the operator $\mathcal{H}_\varepsilon$ in a neighbourhood of the right-hand edge of the gap $(\beta_+ + \alpha_0^2)$.

Theorem 2. *Suppose that assumptions A1) and A3) hold. Then for sufficiently small ε the operator $\mathcal{H}_\varepsilon$ does not have isolated eigenvalues in a neighbourhood of the edge of the gap $\beta_+ + \alpha_0^2$.*

Suppose that assumptions A1) and A2) hold. If $\mu_1 < 0$, then for sufficiently small ε the operator $\mathcal{H}_\varepsilon$ does not have isolated eigenvalues in a neighbourhood of the edge

of the gap $\beta_+ + \alpha_0^2$. If $\mu_1 > 0$, then for sufficiently small ε the operator $\mathcal{H}_\varepsilon$ has a unique isolated eigenvalue λ_ε converging to $\beta_+ + \alpha_0^2$ as $\varepsilon \rightarrow +0$. This eigenvalue is real, simple and has the asymptotics

$$\lambda_\varepsilon = \beta_+ + \alpha_0^2 - e^{-1/(\varepsilon\mu_1)}(e^{\mu_2/\mu_1^2} + O(\varepsilon)), \quad \varepsilon \rightarrow +0. \quad (2.14)$$

Similar results are also true for the left-hand edge of the gap. An analogue of assumption A1) here is as follows.

A4) The edge of the gap β_- is a maximum of only one band function E_m . This maximum is nondegenerate and is attained at an interior point τ_0 of the Brillouin zone $\square^*$. The eigenvalue $E_m(\tau_0)$ is a simple eigenvalue of the operator $\mathcal{H}_0(\tau_0)$. The corresponding eigenfunction ϕ_0 does not vanish identically on $\partial\square$.

Assumptions A2) and A3) are unchanged, except for the replacement of β_+ by β_- throughout. The analogue of formula (2.7) takes the form

$$E_m(\tau) = \beta_- - E_m''(\tau - \tau_0) + O(|\tau - \tau_0|^3), \quad (2.15)$$

and the analogue of d_+ is defined in the same fashion but using the quadratic form E_m in (2.15). The formulae for the quantities μ_1 and μ_2 change sign and are expressed in terms of d_- as follows:

$$\begin{aligned} \mu_1 &:= \frac{d_-}{4\pi^2} \int_{\mathbb{R}^2} V_1 |\psi_0|^2 dx', \\ \mu_2 &:= \frac{d_-}{4\pi^2} \int_{\mathbb{R}^2} \left(V_2 + \frac{V_3}{3} \right) |\psi_0|^2 dx' - \frac{d_-}{4\pi^2} \int_{\mathbb{R}^2} \bar{\psi}_0 V_1 U_0 dx'. \end{aligned}$$

We state an analogue of Theorem 2 for the left-hand edge of the gap.

Theorem 3. *Suppose that assumptions A4) and A3) hold. Then for sufficiently small ε the operator $\mathcal{H}_\varepsilon$ does not have isolated eigenvalues in a neighbourhood of the edge of the gap $\beta_- + \alpha_0^2$.*

Suppose that assumptions A4) and A2) hold. If $\mu_1 < 0$, then for sufficiently small ε the operator $\mathcal{H}_\varepsilon$ does not have isolated eigenvalues in a neighbourhood of the edge of the gap $\beta_- + \alpha_0^2$. If $\mu_1 > 0$, then for sufficiently small ε the operator $\mathcal{H}_\varepsilon$ has a unique isolated eigenvalue λ_ε converging to $\beta_- + \alpha_0^2$ as $\varepsilon \rightarrow +0$. This eigenvalue is real, simple and has the asymptotics

$$\lambda_\varepsilon = \beta_- + \alpha_0^2 + e^{-1/(\varepsilon\mu_1)}(e^{\mu_2/\mu_1^2} + O(\varepsilon)), \quad \varepsilon \rightarrow +0.$$

We now discuss the main results. By Theorem 1, every gap (β_-, β_+) in the spectrum of the operator $\mathcal{H}_0$ for sufficiently small ε generates the gap $(\beta_- + \alpha_0^2, \beta_+ + \alpha_0^2)$ in the essential spectrum of the operator $\mathcal{H}_\varepsilon$, and this gap is independent of ε . In a neighbourhood of the edges of this gap, the operator $\mathcal{H}_\varepsilon$ can have isolated eigenvalues. As Theorems 2 and 3 assert, a necessary condition for the existence of such eigenvalues is the existence of nontrivial bounded solutions of equation (2.8) for the right-hand edge of the gap, and of (2.8) with β_+ replaced by β_- for the left-hand edge of the gap. The existence of a nontrivial solution with the asymptotics (2.9) still does not guarantee the appearance of an eigenvalue—the sign of

the number μ_1 turns out to be important. If the sign is negative, there is still no eigenvalue; it appears only for positive μ_1 . In the latter case, Theorems 2 and 3 describe the basic properties of this eigenvalue. Note that although we do not consider the case $\mu_1 = 0$, the technique we use makes it possible to study this case as well. In this case, the appearance or absence of an eigenvalue is determined by the sign of the number μ_2 , and the exponentially small part of the asymptotics turns out to be of order $O(e^{-1/(\varepsilon^2 \mu_2)})$. Here, the critical case $\mu_2 = 0$ also arises, and we need to calculate further similar numbers according to certain rules, as we do every time the critical situation described above appears, when another constant becomes equal to zero. Conceptually the process of calculating these constants is no different from the process for those presented in the paper, but the calculations turn out to be very cumbersome, and therefore we do not consider this process in the paper. It is also necessary to emphasize that in Theorems 2 and 3 we only consider isolated eigenvalues and do not look at possible eigenvalues embedded in the essential spectrum. The existence of such embedded eigenvalues is not excluded, but our technique does not apply to them, and Theorems 2 and 3 do not give any information about them.

Let us also discuss assumptions A1)–A4). Assumptions A1) and A4) describe the simplest case of the spectral structure of an edge of a gap: a neighbourhood of the boundary is described by only one band function whose corresponding minimum or maximum is nondegenerate, the eigenfunction corresponding to the boundary does not vanish identically on the boundary. This situation is realized, for example, in the case $A_0 = 0$. Here the essential spectrum is the half-axis $[0, +\infty)$ and the lower boundary of the spectrum $\lambda = 0$ is an eigenvalue of the operator $\mathcal{H}_0(\tau)$ for $\tau = 0$; the corresponding eigenfunction is a constant. It is easy to verify that assumption A1) holds here. We also note that assumptions A1) and A4) mean that the band of the essential spectrum adjacent to the edge of the gap under consideration is an interval and does not degenerate to a point. The nondegeneracy of the band of the spectrum we have described here is typical, and can be guaranteed if in addition we assume, for example, that $A_{ij} \in C^2(\mathbb{R}^2)$ (see the general result in [34]).

The situation for assumptions A2) and A3) is similar. We can show that in the general case equation (2.8) has only finitely many nontrivial bounded solutions. One variant of the typical behaviour of such solutions at infinity is described by the asymptotics (2.9). This case is realized in the case $A_0 = 0$, $V_0 = 0$, $\beta_+ = 0$. Here, $\phi_0 = \psi_0 \equiv 1$. In other words, for this case both assumptions A1) and A2) hold simultaneously. In the case $V_0 \neq 0$, both assumption A2) and A3) can be realized. It is sufficient to replace the potential V_0 by tV_0 and to assume that $V_0 \leq 0$. We now let the parameter t increase from 0 to $+\infty$. As is well known, when the parameter passes over certain critical values, eigenvalues of the operator $\mathcal{H}_0 + tV_0$ emerge from the edge of the gap β_+ . This means (see [27]) that for critical values of t equation (2.8) has a nontrivial bounded solution which, in particular, can behave in accordance with (2.9). We also note that the number of critical values of t is countable independently of any concrete choice of the potential V_0 . For the other, noncritical values of t , eigenvalues do not emerge from the edge of the gap, which means there are no nontrivial bounded solutions of equation (2.8).

We also discuss some open questions. A natural continuation of the work in this paper is to study the case of general position without assumptions A1)–A4). Here an edge of a gap can be a point of minimum/maximum for several band functions, the degeneration of the minimum/maximum for every band function is arbitrary, and several eigenfunctions of the operator $\mathcal{H}_0(\tau_0)$ correspond to the edge of the gap. Equation (2.8) can have several nontrivial bounded solutions or none at all. The techniques in this paper can be transferred in part to such a general case, in particular the reduction of the original problem to the study of a regular perturbation of the effective operator (see §4.2). But major changes will be required to analyze the resolvent of the effective operator near the edge of the essential spectrum and for the resulting study of the reduced equation. And as prompted by the analogy with the case of regular perturbations of selfadjoint operators (see [23], [28]), in the general case the asymptotics of the eigenvalues which emerge can not only be exponential but also power-law or logarithmic, that is, we are speaking about modifications of the ansätze.

Another open question is related to the coefficients of the operator. The choice of the functions α and A_{ij} , A_0 described above ensures that ε -independent gaps do exist in the essential in the essential spectrum of the operator $\mathcal{H}_\varepsilon$. But if in the definition of the function α we replaced the number α_0 by some periodic function and/or introduce dependence on the variable x_3 or x_3/ε into the functions A_{ij} , A_0 , then the essential spectrum of the operator $\mathcal{H}_\varepsilon$ would no longer be described by Theorem 1, which is in fact based on the possibility of separating the variables x' and x_3 . Instead, we would have to use the general theory of periodic operators and introduce the corresponding operator on a periodicity cell. The latter operator turns out to be a $\mathcal{PT}$ -symmetric operator with discrete spectrum, and giving a description of the behaviour of such a spectrum turns out to be a very complicated problem. In particular, it is not clear whether the corresponding dispersion laws are real functions or can have complex values. In the latter case the structure of the essential spectrum turns out to be very unusual: bands of the spectrum will be situated in the complex plane. The question of isolated eigenvalues in neighbourhoods of the boundaries of such bands will also require major modification of the techniques in our paper, since we implicitly use the fact that the bands of the essential spectrum are real.

§ 3. The essential spectrum

Here we prove Theorem 1. It follows from (2.2) and (2.6) that the functions V_ε and α_1 are decreasing at least like $O(|x'|^{-1})$ as $x' \rightarrow \infty$. Since the essential spectrum is introduced in terms of characteristic sequences, this spectrum of the operator $\mathcal{H}_\varepsilon$ does not change if we set the functions V_ε and α_1 to be equal to zero. In what follows we assume that this assumption holds for the functions V_ε and α_1 . Then the operator $\mathcal{H}_\varepsilon$ admits a representation in the form of a tensor product $\mathcal{H}_\varepsilon = \mathcal{H}_0 \otimes \mathcal{H}_0^\perp$, where $\mathcal{H}_0^\perp$ is the operator $-d^2/dx_3^2$ on $(0, \varepsilon)$ with the boundary conditions

$$\left(\frac{d}{dx_3} + i\alpha_0 \right) u = 0, \quad x_3 = 0, \quad x_3 = \varepsilon.$$

The spectrum of the operator $\mathcal{H}_0^\perp$ consists of the simple eigenvalues $\alpha_0^2, \pi^2 j^2 / \varepsilon^2$, $j \in \mathbb{N}$ (see [12], §3). Therefore, based on the characteristic sequences of the operator $\mathcal{H}_0$ and the eigenfunctions of the operator $\mathcal{H}_0^\perp$ written out in [12], §3, it is elementary to construct the characteristic sequences of the operator $\mathcal{H}_\varepsilon$ and to prove the embedding

$$\sigma_{\text{ess}}(\mathcal{H}_\varepsilon) \subseteq \sigma_{\text{ess}}(\mathcal{H}_0) + \alpha_0^2 + \bigcup_{j=1}^{\infty} \left(\sigma_{\text{ess}}(\mathcal{H}_0) + \frac{\pi^2 j^2}{\varepsilon^2} \right). \quad (3.1)$$

Further, for $\lambda \notin \sigma_{\text{ess}}(\mathcal{H}_0) + \alpha_0^2 + \bigcup_{j=1}^{\infty} (\sigma_{\text{ess}}(\mathcal{H}_0) + \pi^2 j^2 / \varepsilon^2)$ the resolvent $(\mathcal{H}_\varepsilon - \lambda)^{-1}$ is constructed by a simple separation of the variables x' and x_3 and using a basis of eigenfunctions of the operator $\mathcal{H}_0^\perp$ (see [12], §3, as well as the independent constructions in §4.1). This proves the embedding that is inverse to (3.1).

§ 4. Emergence of eigenvalues

The present section is devoted to the proof of Theorem 2. The proof consists of several stages, each of which is described below in a separate subsection. Throughout the proof, we use $\mathcal{H}$ for short to denote the differential expression

$$\mathcal{H} := - \sum_{i,j=1}^2 \frac{\partial}{\partial x_i} A_{ij} \frac{\partial}{\partial x_j} + A_0,$$

which will be applied to all suitable functions.

4.1. A model periodic problem. In this subsection we consider the auxiliary boundary problem with periodic coefficients

$$\begin{aligned} \left(\mathcal{H} - \frac{\partial^2}{\partial x_3^2} - \alpha_0^2 - \beta_+ + k \right) u &= f \quad \text{in } \Omega_\varepsilon, \\ \left(\frac{\partial}{\partial x_3} + i\alpha_0 \right) u &= 0 \quad \text{on } \partial\Omega_\varepsilon. \end{aligned} \quad (4.1)$$

Here, β_+ is the edge of the gap in the hypothesis of Theorem 2 and f is an arbitrary function in $L_2(\Omega_\varepsilon)$. Let k denote a small complex parameter; we assume that $k \in K_\delta$, where $K_\delta := \{k \in \mathbb{C} : |k| < \delta\} \setminus \mathbb{R}_-$ and δ is a sufficiently small fixed number. We seek a solution of problem (4.1) in the space $W_2^2(\Omega_\varepsilon)$ among the functions satisfying the boundary condition in (4.1). Our main goal is to analyze the solvability of problem (4.1) and determine the nature of the dependence of the solution on ε .

First we point out some facts that follow from [13], §4. We set

$$\begin{aligned} \theta_0(x_3) &:= e^{-i\alpha_0 x_3}, & \theta_p(x_3) &:= \cos \mu_p x_3 - i \frac{\alpha_0}{\mu_p} \sin \mu_p x_3, \\ \varkappa_0 &:= \frac{2i\alpha_0}{1 - e^{-2i\alpha_0 \varepsilon}}, & \varkappa_p &:= \frac{2\mu_p^2}{(\mu_p^2 - \alpha_0^2)\varepsilon}, & \mu_p &:= \frac{\pi p}{\varepsilon}, \quad p \geq 1. \end{aligned}$$

Similarly to Lemma 4.1 in [13], it is easy to prove that any function in $L_2(\Omega_\varepsilon)$ can be represented by a series

$$g(x) = \sum_{p=0}^{\infty} g_p(x') \theta_p(x_3), \quad g_p(x') := \kappa_p \int_0^\varepsilon g(x) \theta_p(x_3) dx_3, \quad (4.2)$$

and satisfies the estimates

$$C_1 \varepsilon^{-2} \|g(x', \cdot)\|_{L_2(0, \varepsilon)}^2 \leq \sum_{p=0}^{\infty} |g_p(x')|^2 \leq C_2 \varepsilon^{-2} \|g(x', \cdot)\|_{L_2(0, \varepsilon)}^2, \quad (4.3)$$

where C_1 and C_2 are constants independent of g , ε and x' . This allows us to separate the variables in problem (4.1):

$$f(x) = \sum_{p=0}^{\infty} f_p(x') \theta_p(x_3), \quad u(x) = \sum_{p=0}^{\infty} u_p(x') \theta_p(x_3),$$

where the u_p are solutions of the equations

$$(\mathcal{H}_0 - \beta_+ + k)u_0 = f_0, \quad \left(\mathcal{H}_0 + \frac{\pi^2 p^2}{\varepsilon^2} - \alpha_0^2 - \beta_+ + k \right) u_p = f_p, \quad p \geq 1. \quad (4.4)$$

The equation for u_0 has a solution, at least for $k \in K_\delta$ for sufficiently small δ . The equations for the u_p have a solution for $k \in \overline{K}_\delta$, for all $p \geq 1$, and for sufficiently small ε , since the distance from the numbers $\pi^2 p^2 / \varepsilon^2$ to the spectrum of the operator $\mathcal{H}_0$ is of order $O(\varepsilon^{-2})$.

Using the standard estimate for the norm of the resolvent of a selfadjoint operator in terms of the distance to its spectrum and the integral equation ($p \geq 1$)

$$\begin{aligned} \sum_{i,j=1}^2 \left(A_{ij} \frac{\partial u_p}{\partial x_j}, \frac{\partial u_p}{\partial x_i} \right)_{L_2(\mathbb{R}^2)} + (A_0 u_p, u_p)_{L_2(\mathbb{R}^2)} \\ + \left(\frac{\pi^2 p^2}{\varepsilon^2} - \alpha_0^2 - \beta_+ + k \right) \|u_p\|_{L_2(\mathbb{R}^2)}^2 = (f_p, u_p)_{L_2(\mathbb{R}^2)}, \end{aligned}$$

it is easy to verify the estimates

$$\|u_p\|_{L_2(\mathbb{R}^2)} \leq \frac{C \varepsilon^2}{p^2} \|f_p\|_{L_2(\mathbb{R}^2)}, \quad \|u_p\|_{W_2^1(\mathbb{R}^2)} \leq \frac{C \varepsilon}{p} \|f_p\|_{L_2(\mathbb{R}^2)},$$

where the constant C is independent of ε , p , and f_p . It follows from these estimates and (4.3) that the series $\sum_{p=1}^{\infty} u_p(x') \theta_p(x_3)$ converges in $W_2^2(\Omega_\varepsilon)$. Moreover, the operator $\mathcal{B}_1^\varepsilon(k): L_2(\Omega_\varepsilon) \rightarrow W_2^2(\Omega_\varepsilon)$, which maps a function f to the sum of this series, satisfies the estimates

$$\|\mathcal{B}_1^\varepsilon(k)\|_{L_2(\Omega_\varepsilon) \rightarrow L_2(\Omega_\varepsilon)} \leq C \varepsilon^2, \quad \|\mathcal{B}_1^\varepsilon(k)\|_{L_2(\Omega_\varepsilon) \rightarrow W_2^1(\Omega_\varepsilon)} \leq C \varepsilon, \quad (4.5)$$

where $\|\cdot\|_{X \rightarrow Y}$ is the norm of a bounded operator from a Banach space X to a Banach space Y , while the constant C is independent of ε . The operator $\mathcal{B}_1^\varepsilon$ is holomorphic for small k . Thus, the solution of problem (4.1) is represented in the form

$$u = \mathcal{B}_0^\varepsilon(k)f + \mathcal{B}_1^\varepsilon(k)f, \quad \mathcal{B}_0^\varepsilon(k)f := u_0 \theta_0, \quad (4.6)$$

where u_0 is a solution of equation (4.4).

4.2. Reducing the equation for the eigenvalues. At this stage our goal is to reduce the eigenvalue problem for the operator $\mathcal{H}_\varepsilon$ to some equation for the operator $\mathcal{H}_0$.

We seek the eigenvalues of the operator $\mathcal{H}_\varepsilon$ near the edge of the gap β_+ in the form $\lambda^\varepsilon = \beta_+ + \alpha_0^2 - k_\varepsilon$, and in the eigenvalue equation $\mathcal{H}_\varepsilon \psi^\varepsilon = \lambda^\varepsilon \psi^\varepsilon$ we make the change of function $\psi^\varepsilon(x) = e^{-i\alpha_1(x)x_3} \phi^\varepsilon(x)$. As a result, for the function ϕ^ε we obtain the boundary problem

$$\begin{aligned} \left(\mathcal{H} - \frac{\partial^2}{\partial x_3^2} - \alpha_0^2 - \beta_+ + k_\varepsilon \right) \phi^\varepsilon &= -\mathcal{L}_1^\varepsilon \phi^\varepsilon \quad \text{in } \Omega_\varepsilon, \\ \left(\frac{\partial}{\partial x_3} + i\alpha_0 \right) \phi^\varepsilon &= 0 \quad \text{on } \partial\Omega_\varepsilon, \end{aligned} \quad (4.7)$$

where $\mathcal{L}_1^\varepsilon$ denotes the differential expression

$$\begin{aligned} \mathcal{L}_1^\varepsilon \phi_\varepsilon &:= ix_3 \sum_{i,j=1}^2 \left(\frac{\partial}{\partial x_i} \left(A_{ij} \frac{\partial \alpha_1}{\partial x_j} \phi^\varepsilon \right) + \frac{\partial \alpha_1}{\partial x_i} A_{ij} \left(\frac{\partial \phi^\varepsilon}{\partial x_j} - ix_3 \frac{\partial \alpha_1}{\partial x_j} \phi^\varepsilon \right) \right) \\ &\quad + 2i\alpha_1 \frac{\partial \phi_\varepsilon}{\partial x_3} + (V_\varepsilon + \alpha_1^2) \phi^\varepsilon. \end{aligned}$$

Problem (4.7) is in fact problem (4.1) if we take $f := -\mathcal{L}_1^\varepsilon \phi^\varepsilon$ as the right-hand side. Then by (4.6) problem (4.7) can be rewritten in the form

$$\phi_\varepsilon = \mathcal{B}_0^\varepsilon(k_\varepsilon)f + \mathcal{B}_1^\varepsilon(k_\varepsilon)f.$$

Substituting this equation into (4.7) we obtain the following equation for f :

$$f = -\mathcal{L}_1^\varepsilon (\mathcal{B}_0^\varepsilon(k_\varepsilon) + \mathcal{B}_1^\varepsilon(k_\varepsilon))f. \quad (4.8)$$

Let $\mathcal{P}_0$ and $\mathcal{P}^\perp$ denote the projection operators that associate the functions g_0 and $\sum_{j=1}^\infty g_p(x')\theta_p(x_3)$ with the function $g \in L_2(\Omega_\varepsilon)$ in the notation in (4.2). The operator $\mathcal{P}_0$ acts from $L_2(\Omega_\varepsilon)$ to $L_2(\mathbb{R}^2)$, and the operator $\mathcal{P}^\perp$ maps to $L_2(\Omega_\varepsilon)$. By (4.3) we have the estimates

$$\|\mathcal{P}_0\|_{L_2(\Omega_\varepsilon) \rightarrow L_2(\mathbb{R}^2)} \leq C\varepsilon^{-1/2}, \quad \|\mathcal{P}^\perp\|_{L_2(\Omega_\varepsilon) \rightarrow L_2(\Omega_\varepsilon)} \leq C. \quad (4.9)$$

We set $f_0 := \mathcal{P}_0 f$ and $f^\perp := \mathcal{P}^\perp f$; then $f = f_0\theta_0 + f^\perp$. We apply the operators $\mathcal{P}_0$ and $\mathcal{P}^\perp$ to equation (4.8) and take account of the obvious equations

$$\mathcal{B}_0^\varepsilon(k)(f_0\theta_0 + f^\perp) = \mathcal{B}_0^\varepsilon(k)(f_0\theta_0), \quad \mathcal{B}_1^\varepsilon(k)(f_0\theta_0 + f^\perp) = \mathcal{B}_1^\varepsilon(k)f^\perp.$$

We see that equation (4.8) is equivalent to the following pair of equations:

$$f_0 = -(V_4 - q_1(\varepsilon)V_3)u_0 - q_2(\varepsilon)\mathcal{L}_2 u_0 - \mathcal{P}_0 \mathcal{L}_1^\varepsilon \mathcal{B}_1^\varepsilon(k_\varepsilon)f^\perp, \quad (4.10)$$

$$f^\perp = -\mathcal{P}^\perp \mathcal{L}_1^\varepsilon \mathcal{B}_1^\varepsilon(k_\varepsilon)f^\perp + \mathcal{L}_3 u_0, \quad (4.11)$$

where we have set

$$\begin{aligned} V_4 &:= \mathcal{P}_0(V_\varepsilon \theta_0) + 2\alpha_0 \alpha_1 + \alpha_1^2, \\ q_1(\varepsilon) &:= \frac{1 - e^{-2i\alpha_0 \varepsilon}(1 + 2i\alpha_0 \varepsilon - 2\alpha_0^2 \varepsilon^2)}{2\alpha_0^2(1 - e^{-2i\alpha_0 \varepsilon})}, \\ q_2(\varepsilon) &:= \frac{1 - (1 + 2i\alpha_0 \varepsilon)e^{-2i\alpha_0 \varepsilon}}{2\alpha_0(1 - e^{-2i\alpha_0 \varepsilon})}, \\ \mathcal{L}_2 u^0 &:= \sum_{i,j=1}^2 \left(\frac{\partial}{\partial x_i} \left(A_{ij} \frac{\partial \alpha_1}{\partial x_j} u^0 \right) + \frac{\partial \alpha_1}{\partial x_i} A_{ij} \frac{\partial u^0}{\partial x_j} \right), \\ \mathcal{L}_3 &:= \mathcal{P}^\perp(x_3 \theta_0)(-i\mathcal{L}_2 + V_3) - \mathcal{P}^\perp(V_\varepsilon \theta_0). \end{aligned}$$

The estimate (4.9) for $\mathcal{P}^\perp$, with (4.5) and the definition of the operator $\mathcal{L}_1^\varepsilon$ imply that

$$\|\mathcal{P}^\perp \mathcal{L}_1^\varepsilon \mathcal{B}_1^\varepsilon(k_\varepsilon)\|_{L_2(\Omega_\varepsilon) \rightarrow L_2(\Omega_\varepsilon)} \leq C\varepsilon,$$

where the constant C is independent of ε and k . Hence, the operator $I + \mathcal{P}^\perp \mathcal{L}_1^\varepsilon \mathcal{B}_1^\varepsilon(k)$ is boundedly invertible in $L_2(\Omega_\varepsilon)$ for sufficiently small ε , which makes it possible to solve equation (4.11):

$$f^\perp = (I + \mathcal{P}^\perp \mathcal{L}_1^\varepsilon \mathcal{B}_1^\varepsilon(k_\varepsilon))^{-1} \mathcal{L}_3 u_0.$$

Substituting this equation into (4.10) and taking the first equation in (4.4) into account, we arrive at the following equation for u_0 :

$$(\mathcal{H}_0 + V_0 - \beta_+ + k_\varepsilon)u_0 = \mathcal{L}_4^\varepsilon(k_\varepsilon)u_0, \quad (4.12)$$

where

$$\mathcal{L}_4^\varepsilon(k) := V_4 - V_0 - q_1(\varepsilon)V_3 - q_2(\varepsilon)\mathcal{L}_2 - \mathcal{P}_0 \mathcal{L}_1^\varepsilon \mathcal{B}_1^\varepsilon(k)(I + \mathcal{P}^\perp \mathcal{L}_1^\varepsilon \mathcal{B}_1^\varepsilon(k))^{-1} \mathcal{L}_3. \quad (4.13)$$

We point out that by the definition of the potentials V_4 and V_0 we have

$$\begin{aligned} V_4(x', \varepsilon) - V_0(x') &= \varkappa_0 \int_0^\varepsilon e^{-2i\alpha_0 x_3} V\left(x', \frac{x_3}{\varepsilon}\right) dx_3 - \int_0^1 V(x', \xi) d\xi \\ &= \int_0^1 (\varepsilon \varkappa_0 e^{-2i\varepsilon \alpha_0 \xi} - 1) V(x', \xi) d\xi, \end{aligned}$$

and so, in particular, the following representation follows:

$$V_4(x', \varepsilon) - V_0(x') = \varepsilon V_1(x') + \varepsilon^2 V_2(x') + \varepsilon^3 V_5(x', \varepsilon), \quad (4.14)$$

$$V_1(x') := -i\alpha_0 \int_0^1 (2\xi - 1) V(x', \xi) d\xi,$$

$$V_2(x') := -\alpha_0^2 \int_0^1 \left(2\xi^2 - 2\xi + \frac{1}{3} \right) V(x', \xi) d\xi,$$

and the potentials $V_i \in C(\mathbb{R}^2)$ satisfy the following estimate, which is uniform in x' and ε and is similar to (2.2):

$$|V_i(x', \varepsilon)| \leq C \chi_1(x') \chi_2(x'). \quad (4.15)$$

Using condition (2.5), it is easy to verify that the potentials V_1 and V_2 introduced in (4.14) can also be defined by formulae (2.10) and (2.12).

Thus, the original eigenvalue problem for the operator $\mathcal{H}_\varepsilon$ is reduced to the equivalent equation (4.12). We have to find values of k_ε such that this equation has nontrivial solutions.

4.3. Analysis of the reduced equation. In this subsection we will complete the proof of Theorem 2. Namely, we describe the values of k_ε for which equation (4.12) has a nontrivial solution, as well as the situations where there are no such values.

A key role in the analysis of equation (4.12) is played by the following assertion.

Lemma 1. *Suppose that assumption A1) holds. For an arbitrary function $F \in L_2(\mathbb{R}^2)$ and an arbitrary $k \in K_\delta$, for a sufficiently small δ the equation*

$$(\mathcal{H}_0 + V_0 - \beta_+ + k)u_0 = \chi_2 F \quad (4.16)$$

has a unique solution in $W_2^2(\mathbb{R}^2)$.

Suppose that assumptions A1) and A2) hold. Then the resolvent

$$\mathcal{A}_1(k)F := (\mathcal{H}_0 + V_0 - \beta_+ + k)^{-1}\chi_2 F \quad (4.17)$$

satisfies the representation

$$\mathcal{A}_1(k)F = -\mathcal{M}_1(F) \ln k \psi_0 + \mathcal{A}_2 F + k^{1/2} \ln k \mathcal{A}_3(k)F, \quad (4.18)$$

$$\mathcal{M}_1(F) := -\frac{d_+}{4\pi^2} (F, \chi_2 \psi_0)_{L_2(\mathbb{R}^2)}. \quad (4.19)$$

Here, $\mathcal{A}_2$ denotes the operator mapping F to the solution of the equation

$$(\mathcal{H} + V_0 - \beta_+)u = \chi_2 F \quad \text{in } \mathbb{R}^2,$$

which can be represented in the form of the sum

$$u(x') = u_1(x') + u_2(x'), \quad u_2 \in W_2^2(\mathbb{R}^2), \quad (4.20)$$

where the function $u_1 \in W_{2,\text{loc}}^2(\mathbb{R}^2)$ for $|x'| > 1$ is given by

$$u_1(x) = 2\pi \mathcal{M}_1(F) \widehat{\phi}_0(x') (-\ln |x'| + c_1) + |x'|^{-2} (x_1 \widehat{\phi}_1(x') + x_2 \widehat{\phi}_1(x')), \quad (4.21)$$

where $\widehat{\phi}_i \in W_{2,\text{per}}^2(\square)$, $i = 1, 2$, are some functions. The estimates

$$\|u_2\|_{W_2^2(\mathbb{R}^2)} + \|\widehat{\phi}_1\|_{W_{2,\text{per}}^2(\square)} + \|\widehat{\phi}_2\|_{W_{2,\text{per}}^2(\square)} \leq C \|F\|_{L_2(\mathbb{R}^2)} \quad (4.22)$$

hold, where the constant C is independent of F . We let $\mathcal{A}_3$ denote an operator from $L_2(\mathbb{R}^2)$ to $W_2^2(\mathbb{R}^2, \chi_1^2 dx')$ that is holomorphic and uniformly bounded in $k \in K_\delta$. The operator $\mathcal{A}_2$ is bounded as an operator from $L_2(\mathbb{R}^2)$ to $W_2^2(\mathbb{R}^2, \chi_1^2 dx')$.

Suppose that assumptions A1) and A3) hold. Then the resolvent (4.17) is uniformly bounded in $k \in K_\delta$ as an operator acting from $L_2(\mathbb{R}^2)$ to $W_2^2(\mathbb{R}^2, \chi_1^2 dx')$.

The proof of this lemma is given in § 5.

We point out that in view of the estimates (2.2) and (2.6) the potential V_0 satisfies the representation

$$V_0 = \chi_1 \chi_2 \tilde{V}_0, \quad \tilde{V}_0 \in C(\mathbb{R}^2) \cap L_\infty(\mathbb{R}^2). \quad (4.23)$$

Moreover, this representation, the estimates (2.6) and (4.15), and the definition (4.13) of the operator $\mathcal{L}_4$ imply that this operator can be represented in the form

$$\mathcal{L}_4^\varepsilon(k) = \chi_2 \mathcal{L}_5^\varepsilon(k), \quad (4.24)$$

where $\mathcal{L}_5^\varepsilon(k)$ is an operator from $W_2^2(\mathbb{R}^2, \chi_1^2 dx')$ to $L_2(\mathbb{R}^2)$ that is holomorphic in $k \in K_\delta$, and the norm of this operator tends to zero as $\varepsilon \rightarrow +0$ uniformly in $k \in K_\delta$.

To investigate equation (4.12) we use the approach given in [26] as it was realized in [27]. First we consider the case when A1) and A2) hold. We substitute the representation (4.24) into (4.12); then by Lemma 1 this equation turns out to be equivalent to the following:

$$\begin{aligned} h_\varepsilon &= \mathcal{L}_5^\varepsilon(k_\varepsilon) \mathcal{A}_1(k) h_\varepsilon, \quad h_\varepsilon := \mathcal{L}_5^\varepsilon(k_\varepsilon) u_0, \\ h_\varepsilon &:= -\mathcal{M}_1(h_\varepsilon) \ln k_\varepsilon \mathcal{L}_5^\varepsilon(k_\varepsilon) \psi_0 + \mathcal{L}_5^\varepsilon(k_\varepsilon) \mathcal{A}_2 h_\varepsilon + k_\varepsilon^{1/2} \ln k_\varepsilon \mathcal{A}_3(k_\varepsilon) h_\varepsilon. \end{aligned} \quad (4.25)$$

It follows from the properties of the operators $\mathcal{L}_5^\varepsilon$, $\mathcal{A}_2$, $\mathcal{A}_3$ that the operator

$$\mathcal{A}_4^\varepsilon(k) := (I + \mathcal{L}_5^\varepsilon(k) \mathcal{A}_2 + k^{1/2} \ln k \mathcal{L}_5^\varepsilon(k) \mathcal{A}_3(k))^{-1}$$

is well defined and is uniformly bounded in ε and $k \in K_\delta$, and is holomorphic in $k \in K_\delta$. We apply this operator to equation (4.25):

$$h_\varepsilon = -\mathcal{M}_1(h_\varepsilon) \ln k_\varepsilon \mathcal{A}_4^\varepsilon(k_\varepsilon) \mathcal{L}_5^\varepsilon(k_\varepsilon) \psi_0.$$

We apply the functional $\mathcal{M}_1$ to the resulting equation and arrive at an equation for k_ε :

$$-\frac{1}{\ln k_\varepsilon} = \mathcal{M}_1(\mathcal{A}_4^\varepsilon(k_\varepsilon) \mathcal{L}_5^\varepsilon(k_\varepsilon) \psi_0). \quad (4.26)$$

This equation describes the values k_ε for which equation (4.12) has a nontrivial solution. This solution can be reconstructed by the formula

$$u_0 = \mathcal{A}_1(k_\varepsilon) \mathcal{A}_4^\varepsilon(k_\varepsilon) \mathcal{L}_5^\varepsilon(k_\varepsilon) \psi_0.$$

This also implies that to every solution k_ε of equation (4.26) there corresponds only one nontrivial solution of equation (4.12).

We now analyze the solvability of equation (4.26). To do this we determine the asymptotics of the right-hand side of this equation for small ε . We expand the operator $\mathcal{A}_4$ into a Neumann series and then use the definition of $\mathcal{M}_1$ and that of the operator $\mathcal{L}_4^\varepsilon$. As a result we obtain

$$\begin{aligned} \mathcal{M}_1(\mathcal{A}_4^\varepsilon(k) \mathcal{L}_5^\varepsilon(k) \psi_0) &= \mathcal{M}_1(\mathcal{L}_5^\varepsilon(k) \psi_0) - \mathcal{M}_1(\mathcal{L}_5^\varepsilon(k) \mathcal{A}_2 \mathcal{L}_5^\varepsilon(k) \psi_0) \\ &\quad + O(k^{1/2} |\ln k| \|\mathcal{L}_5^\varepsilon(k) \psi_0\|_{L_2(\mathbb{R}^2)}) \\ &\quad + \|\mathcal{L}_5^\varepsilon(k) \psi_0\|_{L_2(\mathbb{R}^2)} \|\mathcal{L}_5^\varepsilon(k)\|_{W_2^2(\mathbb{R}^2, \chi_1^2 dx') \rightarrow L_2(\mathbb{R}^2)}^2. \end{aligned} \quad (4.27)$$

Direct calculations show that

$$q_1(\varepsilon) = -\frac{1}{3}\varepsilon^2 + O(\varepsilon^3), \quad q_2(\varepsilon) = \frac{i}{2}\varepsilon + \frac{\alpha_0}{6}\varepsilon^2 + O(\varepsilon^3), \quad (4.28)$$

$$\|\mathcal{P}^\perp(x_3\theta_0)(-i\mathcal{L}_2 + V^\perp)\psi_0\|_{L_2(\Omega_\varepsilon)} = O(\varepsilon^{3/2}), \quad \|\mathcal{P}^\perp(V_\varepsilon\theta_0)\|_{L_2(\Omega_\varepsilon)} = O(\varepsilon^{1/2}). \quad (4.29)$$

From these relations and relations (4.5), (4.9) and (4.14) we derive

$$\begin{aligned} \mathcal{M}_1(\mathcal{L}_5^\varepsilon(k)\psi_0) &= -\frac{d_+}{4\pi^2} \int_{\mathbb{R}^2} \overline{\psi}_0 \mathcal{L}_4^\varepsilon(k)\psi_0 dx' \\ &= -\frac{d_+\varepsilon}{4\pi^2} \int_{\mathbb{R}^2} \overline{\psi}_0 \left(-\frac{i}{2}\mathcal{L}_2 + V_1\right)\psi_0 dx' + O(\varepsilon^2), \end{aligned} \quad (4.30)$$

$$\begin{aligned} \|\mathcal{L}_5^\varepsilon(k)\psi_0\|_{L_2(\mathbb{R}^2)} &= O(\varepsilon), \quad \|\mathcal{L}_5^\varepsilon(k)\|_{W_2^2(\mathbb{R}^2, \chi_1^2 dx') \rightarrow L_2(\mathbb{R}^2)} = O(\varepsilon), \\ \mathcal{M}_1(\mathcal{A}_5^\varepsilon(k)\mathcal{L}_5^\varepsilon(k)\psi_0) &= O(\varepsilon^2), \end{aligned} \quad (4.31)$$

where all the estimates are uniform in $k \in K_\delta$.

We now calculate the integral on the right-hand side of (4.30). First we point out that $\mathcal{L}_2\psi_0 = (\mathcal{H} + V_0 - \beta_+)\alpha_1\psi_0$, and therefore, integrating by parts and taking account of estimates (2.6), we obtain

$$\begin{aligned} \int_{\mathbb{R}^2} \overline{\psi}_0 \mathcal{L}_2\psi_0 dx' &= \lim_{R \rightarrow +\infty} \int_{|x'| < R} \overline{\psi}_0 \mathcal{L}_2\psi_0 dx' \\ &= \lim_{R \rightarrow +\infty} \int_{|x'| < R} \alpha_1\psi_0(\mathcal{H} + V_0 - \beta_+)\overline{\psi}_0 dx' = 0. \end{aligned} \quad (4.32)$$

Thus, it follows from (4.27), (4.30) and (4.31) that

$$\mathcal{M}_1(\mathcal{A}_4^\varepsilon(k)\mathcal{L}_5^\varepsilon(k)\psi_0) = \varepsilon\mu_1 + O(\varepsilon^2) \quad (4.33)$$

uniformly for $k \in K_\delta$, where μ_1 is defined in (2.12). By assumption, $\mu_1 \neq 0$. Therefore by (4.33) equation (4.26) can be solvable in K_δ only for $\mu_1 > 0$, and for small ε the solution has the form

$$k_\varepsilon = e^{-1/(\varepsilon\mu_1)}z_\varepsilon, \quad 0 < C_1 \leq |z_\varepsilon| \leq C_2, \quad (4.34)$$

where C_1 and C_2 are some constants independent of ε .

To analyze the solvability of equation (4.26) we need to make the asymptotics of the right-hand side of this equation as $\varepsilon \rightarrow +0$ more precise. This is given by the following lemma.

Lemma 2. *For any root $k_\varepsilon \in K_\delta$ of equation (4.26)*

$$\mathcal{M}_1(\mathcal{A}_4^\varepsilon(k_\varepsilon)\mathcal{L}_5^\varepsilon(k_\varepsilon)\psi_0) = \varepsilon\mu_1 + \varepsilon^2\mu_2 + O(\varepsilon^3),$$

where μ_1, μ_2 are defined by formulae (2.12), (2.13).

The proof of this lemma is given in § 5.

Lemma 2 and equation (4.26) make it possible to refine the asymptotics (4.34):

$$k_\varepsilon = e^{-1/(\varepsilon\mu_1)}(e^{\mu_2/\mu_1^2} + O(\varepsilon)), \quad \varepsilon \rightarrow +0. \quad (4.35)$$

These a priori asymptotics for the roots of equations (4.26) makes it possible to seek these roots in the form (4.34), where $z_\varepsilon \rightarrow e^{\mu_2/\mu_1^2}$ as $\varepsilon \rightarrow +0$. Equation (4.26) rewritten for z_ε can easily be analyzed if we repeat almost word-for-word the proof of Theorem 1.7 in [27]. As a result we see that equation (4.26) has a unique solution and this solution satisfies the asymptotics (4.35). This means that under the condition $\mu_1 > 0$ the operator $\mathcal{H}_\varepsilon$ has a unique isolated eigenvalue $\lambda_\varepsilon = \beta_+ + \alpha_0^2 - k_\varepsilon$ near the edge of the essential spectrum β_+ . This eigenvalue is simple and satisfies the asymptotics (2.14). Since eigenvalues of $\mathcal{P}$ -pseudo-Hermitian operators are complex conjugates, the uniqueness of the eigenvalue λ_ε implies that it is real.

Now suppose that assumptions A1) and A3) hold. In this case equation (4.25) has to be replaced by

$$h_\varepsilon = \mathcal{L}_5^\varepsilon(k_\varepsilon)(\mathcal{H}_0 + V_0 - \beta_+ + k_\varepsilon)^{-1}h_\varepsilon.$$

And since here the resolvent is uniformly bounded in k_ε by Lemma 1, while the norm of the operator $\mathcal{L}_5^\varepsilon$ tends to zero as $\varepsilon \rightarrow +0$, this equation only has the trivial solution. This proves that there are no isolated eigenvalues of the operator $\mathcal{H}_\varepsilon$ in K_δ for small ε .

§ 5. Auxiliary lemmas

In this section we prove the auxiliary lemmas that were used in the preceding section. We begin with some preliminary results and assertions.

In Lemma 4.5 of [27] it was shown that the operator $(\mathcal{H}_0 - \beta_+ + k)^{-1}\chi_2$ acting from $L_2(\mathbb{R}^2)$ to $W_2^2(\mathbb{R}^2)$ is well defined for $k \in K_\delta$ and has the representation

$$\begin{aligned} (\mathcal{H}_0 - \beta_+ + k)^{-1}\chi_2 g &= -\mathcal{M}_2(g)\widehat{\phi}_0 \ln k + \mathcal{A}_5 + k^{1/2}\mathcal{A}_6(k), \\ \mathcal{M}_2(g) &:= -\frac{d_+}{4\pi^2}(g, \chi_2\widehat{\phi}_0)_{L_2(\mathbb{R}^2)}, \quad \widehat{\phi}_0(x') := e^{i\tau_0 \cdot x'}\phi_0(x'), \end{aligned} \quad (5.1)$$

where the operator $\mathcal{A}_5: L_2(\mathbb{R}^2) \rightarrow W_2^2(\mathbb{R}^2, \chi_1^2 dx')$ is bounded, while the operator $\mathcal{A}_6(k): L_2(\mathbb{R}^2) \rightarrow W_2^2(\mathbb{R}^2, \chi_1^2 dx')$ is uniformly bounded and holomorphic for $k \in K_\delta$. Here, $W_2^2(\mathbb{R}^2, \chi_1^2 dx')$ is the weighted Sobolev space of functions in $W_{2,\text{loc}}^2(\mathbb{R}^2)$ with finite norm

$$\|u\|_{W_2^2(\mathbb{R}^2, \chi_1^2 dx')}^2 := \sum_{i,j=1}^2 \left\| \chi_1 \frac{\partial^2 u}{\partial x_i \partial x_j} \right\|_{L_2(\mathbb{R}^2)}^2 + \sum_{i=1}^2 \left\| \chi_1 \frac{\partial u}{\partial x_i} \right\|_{L_2(\mathbb{R}^2)}^2 + \|\chi_1 u\|_{L_2(\mathbb{R}^2)}^2.$$

In what follows we shall need some additional properties of the operator $\mathcal{A}_5$, which can be conveniently stated in the form of the following lemma.

Lemma 3. *For an arbitrary function $g \in L_2(\mathbb{R}^2)$ the function $u = \mathcal{A}_5 g$ is a solution of the equation*

$$(\mathcal{H} - \beta_+)v = \chi_2 g \quad \text{in } \mathbb{R}^2, \quad (5.2)$$

which can be represented in the form (4.20), where the function $u_1 \in W_{2,\text{loc}}^2(\mathbb{R}^2)$ for $|x'| > 1$ satisfies formula (4.21) and estimates (4.20) with $\mathcal{M}_1(F)$ replaced by $\mathcal{M}_2(g)$, and F replaced by g .

Proof. For an arbitrary function $g \in L_2(\mathbb{R}^2)$, the function $u = (\mathcal{H}_0 - \beta_+ + k)^{-1} \chi_2 g$ is a solution of the equation

$$(\mathcal{H}_0 - \beta_+ + k)u = \chi_2 g.$$

We substitute the right-hand side of the representation (5.1) for u and then take the inner product in $L_2(\mathbb{R}^2)$ of the resulting equation with an arbitrary function $\xi \in C_0^\infty(\mathbb{R}^2)$. Then, taking into account the aforementioned properties of the operators $\mathcal{A}_5$ and $\mathcal{A}_6$, we pass to the limit as $k \rightarrow 0$. As a result we obtain

$$\int_{\mathbb{R}^2} \bar{\xi} (\mathcal{H} - \beta_+) \mathcal{A}_5 g \, dx' = \int_{\mathbb{R}^2} \bar{\xi} \chi_2 g \, dx'.$$

Since the function ξ is arbitrary, this means that the function $v := \mathcal{A}_5 g$ is a solution of equation (5.2).

We now look at the behaviour of the solution of this equation at infinity. Let $\Phi(x', \tau)$ be an eigenfunction of the operator $\mathcal{H}_0(\tau_0)$ corresponding to $E_m(\tau)$. By assumption A1), the functions Φ and E_m are holomorphic in τ at least in a small fixed neighbourhood of τ_0 . The holomorphicity of Φ is understood in the sense of the norm of $W_2^2(\square)$.

In view of the results of [27], §4, the following representation holds for the operator $\mathcal{A}_5$:

$$\mathcal{A}_5 = \mathcal{A}_7 + \mathcal{A}_8 + \mathcal{A}_9, \quad (5.3)$$

where $\mathcal{A}_9$ is some bounded operator from $L_2(\mathbb{R}^2)$ to $W_2^2(\mathbb{R}^2)$, while the operators $\mathcal{A}_i: L_2(\mathbb{R}^2) \rightarrow W_2^2(\mathbb{R}^2, \chi_1^2 dx')$, $i = 7, 8$, are defined by

$$\begin{aligned} (\mathcal{A}_7 g)(x') &:= \frac{1}{4\pi^2} \int_{E_m''(\tau - \tau_0) < \eta^2} \left(\frac{e^{i\tau \cdot x'} \Phi(x', \tau) \widehat{g}(\tau) \zeta(\tau)}{E_m(\tau) - \beta_+} - \frac{\widehat{\phi}_0(x') \widehat{g}(\tau_0)}{E_m''(\tau - \tau_0)} \right) d\tau \\ &\quad + \frac{d_+}{2\pi} \widehat{\phi}_0(x') \widehat{g}(\tau_0) \ln \eta, \\ (\mathcal{A}_8 g)(x') &:= \frac{1}{4\pi^2} \int_{\square^*} \frac{e^{i\tau \cdot x'} \Phi(x', \tau) \widehat{g}(\tau) (1 - \zeta(\tau))}{E_m(\tau) - \beta_+} d\tau. \end{aligned}$$

Here, η is some small fixed number, $\zeta(\tau)$ is an infinitely differentiable cut-off function that is equal to 1 in the domain $\{\tau: E_m''(\tau - \tau_0) < \eta^2/2\}$, and equal to zero outside $\{\tau: E_m''(\tau - \tau_0) > \eta^2\}$, and

$$\widehat{g}(\tau) := \int_{\mathbb{R}^2} \overline{\Phi(y, \tau)} e^{-i\tau y} \chi_2(y) g(y) \, dy.$$

By assumption (2.4) about the function χ_2 , the function $\widehat{g}$ is twice differentiable in τ in $\square^*$ and we have the estimates

$$|\widehat{g}| \leq C, \quad \left| \frac{\partial \widehat{g}}{\partial \tau_i} \right| \leq C, \quad \left| \frac{\partial^2 \widehat{g}}{\partial \tau_i \partial \tau_j} \right| \leq C,$$

where the constant C is independent of τ .

Since the function ζ vanishes identically in a neighbourhood of the point τ_0 , it follows that the integrand in the definition of the operator $\mathcal{A}_8$ has no singularities.

Therefore in view of Theorem 2.2.5 in Ch. 2 of [33] the operator $\mathcal{A}_8$ acts from $L_2(\mathbb{R}^2)$ to $W_2^2(\mathbb{R}^2)$ and is bounded.

Due to the properties of the functions Φ and $\hat{g}$ described above, the integrand in the definition of the operator $\mathcal{A}_7$ can be represented as follows:

$$\begin{aligned} & \left(\frac{e^{i\tau \cdot x'} \Phi(x', \tau) \hat{g}(\tau) \zeta(\tau)}{E_m(\tau) - \beta_+} - \frac{\phi_0(x) \hat{g}(\tau_0)}{E_m''(\tau - \tau_0)} \right) = \hat{\Phi}_0(x) \frac{e^{i(\tau - \tau_0) \cdot x'} - 1}{E_m''(\tau - \tau_0)} \\ & + e^{i\tau \cdot x'} \frac{(\tau_1 - \tau_1^0) \hat{\Phi}_1(x') + (\tau_2 - \tau_2^0) \hat{\Phi}_2(x')}{E_m''(\tau - \tau_0)} + e^{i\tau \cdot x'} \hat{\Phi}_3(x', \tau), \end{aligned}$$

where $\hat{\Phi}_0(x') := \hat{g}(\tau_0) \hat{\phi}_0(x')$, $\hat{\Phi}_i \in W_{2,\text{per}}^2(\square)$, $i = 1, 2, 3$, and the following estimates hold, which are uniform in τ and g :

$$\sum_{i=1}^3 \|\hat{\Phi}_i\|_{W_2^2(\square)} \leq C \|g\|_{L_2(\mathbb{R}^2)}. \quad (5.4)$$

Consequently, the operator $\mathcal{A}_7$ can be represented as the sum

$$\begin{aligned} \mathcal{A}_7 &= \mathcal{A}_{10} + \mathcal{A}_{11} + \mathcal{A}_{12}, \quad (5.5) \\ (\mathcal{A}_{10}g)(x') &:= \frac{1}{4\pi^2} \hat{\Phi}_0(x') \int_{E_m''(\tau - \tau_0) < \eta^2} \frac{e^{i(\tau - \tau_0) \cdot x'} - 1}{E_m''(\tau - \tau_0)} d\tau + \frac{d_+}{2\pi} \hat{\Phi}_0(x') \ln \eta, \\ (\mathcal{A}_{11}g)(x') &:= \frac{1}{4\pi^2} e^{i\tau_0 \cdot x'} \int_{E_m''(\tau - \tau_0) < \eta^2} \frac{e^{i(\tau - \tau_0) \cdot x'} (\tau_1 - \tau_1^0) \hat{\Phi}_1(x') + (\tau_2 - \tau_2^0) \hat{\Phi}_2(x')}{E_m''(\tau - \tau_0)} d\tau, \\ (\mathcal{A}_{12}g)(x') &:= \frac{1}{4\pi^2} \int_{E_m''(\tau - \tau_0) < \eta^2} e^{i\tau \cdot x'} \hat{\Phi}_3(x', \tau) d\tau. \end{aligned}$$

The integral in the definition of the operator $\mathcal{A}_{12}$ can be understood to be taken over the entire Brillouin zone if we extend $\hat{\Phi}_3$ by zero outside $\{\tau : E_m''(\tau - \tau_0) < \eta^2\}$. Therefore by estimate (5.4) and Theorem 2.2.5 in Ch. 2 of [33] the operator $\mathcal{A}_{12}$ is a bounded operator from $L_2(\mathbb{R}^2)$ to $W_2^2(\mathbb{R}^2)$.

We now analyze the operator $\mathcal{A}_{10}$. Clearly, there exists a linear change of variables $\tau - \tau_0 = Bz$, where B is a constant symmetric 2×2 matrix such that $E_m''(\tau - \tau_0) = |z|^2$ and $\det B = d_+$. We make this change of variables in the integral for the operator $\mathcal{A}_{10}$:

$$(\mathcal{A}_{10}g)(x') := \frac{d_+}{2\pi} \hat{\Phi}_0(x') \left(\frac{1}{2\pi} \int_{|z| < \eta} \frac{e^{iz \cdot Bx'} - 1}{|z|^2} dz + \ln \eta \right).$$

To calculate the resulting integral we pass to polar coordinates (ρ, φ) corresponding to the variables ζ , and to polar coordinates (r, ς) corresponding to the variables Bx' . As a result we obtain

$$\begin{aligned} \frac{1}{2\pi} \int_{|z| < \eta} \frac{e^{iz \cdot Bx'} - 1}{|z|^2} dz + \ln \eta &= \frac{1}{2\pi} \int_0^\eta \frac{d\rho}{\rho} \int_0^{2\pi} \frac{e^{i\rho \cos(\varphi - \varsigma)} - 1}{\rho} d\varphi + \ln \eta \\ &= \int_0^\eta \frac{J_0(r\rho) - 1}{\rho} d\rho + \ln \eta = \int_0^{\eta r} \frac{J_0(t) - 1}{t} dt + \ln \eta \\ &= -\ln r + c_1 - \int_{\eta r}^{+\infty} \frac{J_0(t)}{t} dt, \quad \int_{\eta r}^{+\infty} \frac{J_0(t)}{t} dt = O(r^{-3/2}), \quad r \rightarrow +\infty. \end{aligned}$$

Thus, we derive the following asymptotic formula for $\mathcal{A}_{10}g$:

$$(\mathcal{A}_{10}g)(x') = \frac{d_+}{2\pi} \widehat{\Phi}_0(x') (-\ln|x'| + c_1 + O(|x'|^{-3/2})), \quad x' \rightarrow \infty. \quad (5.6)$$

The behaviour of the function $\mathcal{A}_{11}g$ as $x' \rightarrow \infty$ is analyzed in similar fashion. Namely, after passing to polar coordinates in the definition of the operator $\mathcal{A}_{11}$, it is transformed into the form

$$\begin{aligned} (\mathcal{A}_{11}g)(x') &= \frac{d_+}{2\pi} e^{i\tau_0 \cdot x'} \int_0^\eta d\rho \int_0^{2\pi} e^{ir\rho \cos(\varphi-\varsigma)} (\widetilde{\Phi}_1(x') \cos \varphi + \widetilde{\Phi}_2(x') \sin \varphi) d\varphi \\ &= \frac{d_+}{2\pi} e^{i\tau_0 \cdot x'} \int_0^\eta d\rho \int_0^{2\pi} e^{ir\rho \cos \varphi} (\widetilde{\Phi}_1(x') \cos(\varphi + \varsigma) + \widetilde{\Phi}_2(x') \sin(\varphi + \varsigma)) d\varphi, \end{aligned}$$

where the $\widetilde{\Phi}_i$ are linear combinations of the functions $\widehat{\Phi}_i$ with constant coefficients. In view of the formulae

$$\int_0^{2\pi} e^{ir\rho \cos \varphi} \sin \varphi d\varphi = 0, \quad \int_0^{2\pi} e^{ir\rho \cos \varphi} \cos \varphi d\varphi = \frac{J'_0(r\rho)}{ir\rho}$$

we finally obtain

$$\begin{aligned} (\mathcal{A}_{11}g)(x') &= \frac{d_+}{2\pi ir} e^{i\tau_0 \cdot x'} (\widetilde{\Phi}_1(x') \cos \varsigma + \widetilde{\Phi}_2(x') \sin \varsigma) \int_0^{\eta r} \frac{J'_0(t)}{t} dt, \\ \int_0^{\eta r} \frac{J'_0(t)}{t} dt &= \left(\int_0^{+\infty} - \int_{\eta r}^{+\infty} \right) \frac{J'_0(t)}{t} dt = -1 + O(r^{-3/2}), \quad r \rightarrow +\infty. \end{aligned}$$

These relations, relations (5.6), (5.5), (5.3), and the boundedness of the operators $\mathcal{A}_i: L_2(\mathbb{R}^2) \rightarrow W_2^2(\mathbb{R}^2)$, $i = 9, 11$, imply the representation (4.20) and the asymptotics (4.21) both hold. Lemma 3 is proved.

We point out some consequences of the lemma proved above. We represent a nontrivial solution of equation (2.8) with the asymptotics (2.9) in the form $\psi_0 = \widehat{\phi}_0 + \widetilde{\psi}_0$. The function $\widetilde{\psi}_0$ is decreasing at infinity and satisfies the equation

$$(\mathcal{H} - \beta_+) \widetilde{\psi}_0 = -V_0(\widetilde{\psi}_0 + \widehat{\phi}_0). \quad (5.7)$$

In view of the representation (4.20), (4.21) we conclude that

$$\widetilde{\psi}_0 = \mathcal{A}_5 g_0, \quad g_0 := -V_0(\widehat{\phi}_0 + \widetilde{\psi}_0), \quad \mathcal{M}_2(g_0) = 0. \quad (5.8)$$

In addition we point out that when assumption A2) holds, equation (2.8) does not have nontrivial solutions satisfying the representation (4.20), (4.21). Indeed, let u_0 be such a solution with some g . Set $\Pi_N := \{x' \in \mathbb{R}^2: |x_1| < N, |x_2| < N\}$, where $N \in \mathbb{N}$, let (ν_1, ν_2) be the outward normal to $\partial\Pi_N$, and integrate by parts as follows:

$$\begin{aligned} 0 &= \lim_{N \rightarrow +\infty} \int_{\Pi_N} \overline{\psi}_0 (\mathcal{H} + V_0 - \beta_+) u_0 dx' \\ &= \lim_{N \rightarrow +\infty} \sum_{i,j=1}^2 \int_{\partial\Pi_N} \left(-\overline{\psi}_0 A_{ij} \frac{x_i}{R} \frac{\partial u_0}{\partial x_j} - A_{ij} \frac{\partial \overline{\psi}_0}{\partial x_i} \frac{x_j}{R} u_0 \right) dS \\ &= -2\pi \mathcal{M}_2(g) S_0, \quad S_0 := \lim_{N \rightarrow +\infty} \sum_{i,j=1}^2 \int_{\partial\Pi_N} |\phi_0|^2 A_{ij} \nu_i \frac{x_j}{|x'|^2} dS. \end{aligned} \quad (5.9)$$

We now calculate the coefficient S_0 . To do this, we split the integral defining S_0 into four integrals over the sides of the square Π_N and then make the change of variables $x_1 = Ny_1$, $x_2 = Ny_2$ in the resulting integrals. We further use the following easily verifiable formula:

$$\lim_{N \rightarrow +\infty} \int_{-1}^1 \frac{f(tN)t^a dt}{t^2 + 1} = \int_{-1}^1 \frac{t^a dt}{t^2 + 1} \int_0^1 f(z) dz, \quad a = 0, 1,$$

where $f(z)$ is an arbitrary continuous 1-periodic function. As a result we obtain

$$S_0 = 2\pi \int_0^1 (A_{11}(0, z)|\phi_0(0, z)|^2 + A_{22}(z, 0)|\phi_0(z, 0)|^2) dz. \quad (5.10)$$

This equation and the ellipticity condition imply that $S_0 \neq 0$, since otherwise $\phi_0 \equiv 0$ on the boundary of the set $\square$, which contradicts assumption A1). In turn, by (5.9) this means that the coefficient $\mathcal{M}_2(g)$ in (4.21) for u_0 is equal to zero. Therefore the solution u_0 is decreasing at infinity. On the other hand, by assumption A2), equation (2.8) has a unique bounded solution, which is not decaying at infinity.

Proof of Lemma 1. Equation (4.16) is solvable, since $\beta_+ - k$ is not contained in the spectrum of the operator $\mathcal{H}_0$. We set $g := F - \chi_1 \tilde{V}_0 u_0$. Recall that the function $\tilde{V}_0$ was defined in (4.23), and in view of the properties of the function χ_1 (see (2.3)) and as $u_0 \in W_2^2(\mathbb{R}^2)$ we have $g \in L_2(\mathbb{R}^2)$.

Consider the case where assumptions A1) and A2) hold. It follows from equation (4.16) that

$$u_0 = (\mathcal{H}_0 - \beta_+ + k)^{-1} \chi_2 g. \quad (5.11)$$

Substituting this relation and (5.1) into equation (4.16) and then cancelling out χ_2 gives

$$g + \mathcal{M}_2(g) \tilde{V} \chi_1 \hat{\phi}_0 \ln k + \tilde{V}_0 \chi_1 \mathcal{A}_5 g + k^{1/2} \tilde{V}_0 \chi_1 \mathcal{A}_6(k) g = F. \quad (5.12)$$

The properties of the operator $\mathcal{A}_6(k)$ described after formula (5.1) and relations (4.23) imply that the operator $\tilde{V}_0 \chi_1 \mathcal{A}_6(k)$ is uniformly bounded for small $k \in K_\delta$ as an operator in $L_2(\mathbb{R}^2)$. Using (4.23) again and taking account of the representation (4.20)–(4.22), similarly to Lemma 3.2 in [27] it is easy to verify that the operator $\tilde{V}_0 \chi_1 \mathcal{A}_5$ is compact as an operator in $L_2(\mathbb{R}^2)$.

For an arbitrary function $h \in L_2(\mathbb{R}^2)$, consider the equation

$$g + \tilde{V}_0 \chi_1 \mathcal{A}_5 g = h. \quad (5.13)$$

Setting $v := \mathcal{A}_5 g$ we see that this equation is equivalent to the equation

$$(\mathcal{H} + V_0 - \beta_+) v = \chi_2 h \quad \text{in } \mathbb{R}^2.$$

As shown above, this equation with $h = 0$ does not have nontrivial solutions (see the discussion after relations (5.9)). Consequently, equation (5.13) with $h = 0$ does not have nontrivial solutions, which means that, by the compactness of the operator $\tilde{V}_0 \chi_1 \mathcal{A}_5$ and the Fredholm alternatives, the operator $(I + \tilde{V}_0 \chi_1 \mathcal{A}_5)$ is boundedly invertible. In view of the properties of the operator $\tilde{V}_0 \chi_1 \mathcal{A}_6(k)$ described above, this implies the existence and boundedness of the inverse operator

$$\mathcal{A}_{13}(k) := (I + \tilde{V}_0 \chi_1 \mathcal{A}_5 + k^{1/2} \tilde{V}_0 \chi_1 \mathcal{A}_6(k))^{-1}. \quad (5.14)$$

We apply this operator to equation (5.11):

$$g + \mathcal{M}_2(g) \mathcal{A}_{13}(k) \tilde{V}_0 \chi_1 \hat{\phi}_0 \ln k = \mathcal{A}_{13}(k) F. \quad (5.15)$$

We apply the functional $\mathcal{M}_2$ to this equation and obtain an expression for $\mathcal{M}_2(g)$ from the resulting equation:

$$\mathcal{M}_2(g) = \frac{\mathcal{M}_2(\mathcal{A}_{13}(k) F)}{1 + \mathcal{M}_2(\mathcal{A}_{13}(k) \tilde{V}_0 \chi_1 \hat{\phi}_0) \ln k}. \quad (5.16)$$

We now verify that the denominator in this expression does not vanish for sufficiently small k .

First we point out that (5.7), (5.8) and (5.13) imply that the functions

$$\tilde{g} := \mathcal{A}_{13}(0) \tilde{V}_0 \chi_1 \hat{\phi}_0, \quad \tilde{v} := \mathcal{A}_5 g \quad (5.17)$$

generate a solution of equation (5.2) with right-hand side $-V_0 \hat{\phi}_0$, and therefore,

$$\tilde{v} = \tilde{\psi}_0, \quad \tilde{g} = g_0, \quad \mathcal{M}_2(\tilde{g}) = 0. \quad (5.18)$$

From the definition (5.14) of the operator $\mathcal{A}_{13}$ and the definition of the functional $\mathcal{M}_2$ we now derive

$$1 + \mathcal{M}_2(\mathcal{A}_{13}(k) \tilde{V}_0 \chi_1 \hat{\phi}_0) \ln k = 1 + O(k |\ln k|).$$

This proves that (5.16) is well defined and gives the following asymptotic representation:

$$\mathcal{M}_2(g) = \mathcal{M}_2(\mathcal{A}_{13}(0) F) + k \ln k \mathcal{M}_3(k, F), \quad (5.19)$$

where $\mathcal{M}_3(k, \cdot) : L_2(\mathbb{R}^2) \rightarrow \mathbb{C}$ is some functional that is uniformly bounded in $k \in K_\delta$.

We substitute (5.16) into (5.15) and finally find the solution of equation (5.12):

$$g = \mathcal{A}_{13}(k) F - \frac{\mathcal{M}_2(\mathcal{A}_{13}(k) F) \mathcal{A}_{13}(k) \tilde{V}_0 \chi_1 \hat{\phi}_0}{1 + \mathcal{M}_2(\mathcal{A}_{13}(k) \tilde{V}_0 \chi_1 \hat{\phi}_0) \ln k} \ln k. \quad (5.20)$$

In view of (5.19), (5.18) and (5.8) we have

$$g = \mathcal{A}_{13}(0) F - \mathcal{M}_2(\mathcal{A}_{13}(0) F) g_0 \ln k + k \ln^2 k \mathcal{A}_{14}(k) F, \quad (5.21)$$

where $\mathcal{A}_{14}(k)$ is an operator in $L_2(\mathbb{R}^2)$ that is uniformly bounded in $k \in K_\delta$. Together with (5.1), (5.18) and (5.19), this implies the following:

$$u_0 = -\mathcal{M}_2(\mathcal{A}_{13}(0) F) \psi_0 \ln k + \mathcal{A}_5 \mathcal{A}_{13}(0) F + k^{1/2} \ln k \mathcal{A}_{15}(k) F,$$

where $\mathcal{A}_{15}(k)$ is an operator from $L_2(\mathbb{R}^2)$ to $W_2^2(\mathbb{R}^2, \chi_1^2 dx')$ that is uniformly bounded in $k \in K_\delta$.

We calculate $\mathcal{M}_2(\mathcal{A}_{13}(0) F)$. Let $v := \mathcal{A}_5 \mathcal{A}_{13}(0) F$. This function is a solution of equation (5.2) with right-hand side $(\chi_2 F - V_0 v)$ and $\mathcal{A}_{13}(0) F = F - \tilde{V}_0 \chi_1 v$. By the definition of $\mathcal{M}_2$ we have

$$\mathcal{M}_2(\mathcal{A}_{13}(0) F) = -\frac{d_+}{4\pi^2} (F, \chi_2 \hat{\phi}_0)_{L_2(\mathbb{R}^2)} + \frac{d_+}{4\pi^2} (\tilde{V}_0 \chi_1 v, \chi_2 \hat{\phi}_0)_{L_2(\mathbb{R}^2)}. \quad (5.22)$$

By using the definition of the function v and integrating by parts we obtain

$$\begin{aligned} (\widetilde{V}_0 \chi_1 v, \chi_2 \widehat{\phi}_0)_{L_2(\mathbb{R}^2)} &= \int_{\mathbb{R}^2} V_0 v \overline{\widehat{\phi}_0} dx = - \int_{\mathbb{R}^2} v (\mathcal{H} + V_0 - \beta_+) \overline{\widehat{\psi}_0} dx \\ &= - \int_{\mathbb{R}^2} \chi_2 F \overline{\widehat{\psi}_0} dx, \end{aligned}$$

and using this and (5.22) we derive

$$\mathcal{M}_2(\mathcal{A}_{13}(0)F) = (F, \chi_2 \psi_0)_{L_2(\mathbb{R}^2)}.$$

The obtained formula, as well as (5.11), (5.21), (5.7), (5.8), and (5.18) give the representation (4.18).

Now suppose that assumptions A1) and A3) hold. In this case equation (5.12) remains valid. If in addition equation (2.8) does not have logarithmically increasing solutions (see (4.21)), then this means the operator $I + \widetilde{V}_0 \chi_1 \mathcal{A}_5$ is uniquely solvable and the operator $\mathcal{A}_{13}$ in (5.14) exists. The function $\widetilde{g}$ in (5.17) satisfies $\mathcal{M}_2(\widetilde{g}) \neq 0$. Indeed, the function $\psi_0 := \widetilde{v} - \widehat{\phi}_0$, where $\widetilde{v}$ is defined in (5.17), is a nontrivial solution of equation (2.8). In addition, if $\mathcal{M}_2(\widetilde{g}) = 0$, then by Lemma 3 the function ψ_0 is a nontrivial bounded solution of this equation, which does not exist by assumption A3). Further, formula (5.19) should be replaced by

$$\mathcal{M}_2(g) = \frac{\mathcal{M}_2(\mathcal{A}_{13}(0)F)}{1 + \mathcal{M}_2(\widetilde{g}) \ln k} + \mathcal{M}_4(k)F,$$

where the functional $\mathcal{M}_4(k): L_2(\mathbb{R}^2) \rightarrow \mathbb{C}$ is uniformly bounded in $k \in K_\delta$. Formula (5.20) remains valid, while relation (5.21) is replaced by

$$g = \mathcal{A}_{13}(0)F - \frac{\mathcal{M}_2(\mathcal{A}_{13}(0)F) \ln k}{1 + \mathcal{M}_2(\widetilde{g}) \ln k} \widetilde{g} + k \mathcal{A}_{16}(k)F,$$

where $\mathcal{A}_{16}(k)$ is an operator in $L_2(\mathbb{R}^2)$ that is uniformly bounded in $k \in K_\delta$. This fact and (5.1) imply that the resolvent $(\mathcal{H}_0 + V_0 - \beta_+ + k)^{-1}$ is uniformly bounded, as required.

If equation (2.8) has a logarithmically increasing solution, then it is unique; otherwise the difference of such solutions generates a nontrivial bounded solution, which contradicts assumption A3). Let ψ_0 be such an increasing solution. Then $\psi_0 = -\mathcal{A}_5(0)\widehat{g}$, where $\widehat{g} := -\widetilde{V}_0 \chi_1 \psi_0$ and $\mathcal{M}_2(\widehat{g}) \neq 0$. The function $\widehat{g}$ is the unique solution of equation (5.13) with homogeneous right-hand side. The corresponding solution of the conjugate equation is the function $\chi_2 \psi_0$. Indeed, for any $g \in L_2(\mathbb{R}^2)$ we have

$$\begin{aligned} ((I + (\widetilde{V}_0 \chi_1 \mathcal{A}_5)^*) \chi_2 \psi_0, g)_{L_2(\mathbb{R}^2)} &= (\chi_2 \psi_0, (I + \widetilde{V}_0 \chi_1 \mathcal{A}_5)g)_{L_2(\mathbb{R}^2)} \\ &= \int_{\mathbb{R}^2} \overline{\psi_0} (\mathcal{H} + V_0 - \beta_+) \mathcal{A}_5 g dx' = \int_{\mathbb{R}^2} \mathcal{A}_5 g (\mathcal{H} + V_0 - \beta_+) \overline{\psi_0} dx' = 0. \end{aligned}$$

We have to look for a solution of (5.12) in the form $g = \mathcal{M}_5(g_1)\widehat{g} + g_1$, where $\mathcal{M}_5$ is some functional and $(g_1, \widehat{g})_{L_2(\mathbb{R}^2)} = 0$. After substituting this representation

into (5.12), in view of the equation for $\widehat{g}$ we obtain

$$\begin{aligned} g_1 + \widetilde{V}_0 \chi_1 \mathcal{A}_5 g_1 &= -\mathcal{M}_5(g_1) \cdot \mathcal{M}_2(\widehat{g}) \widetilde{V}_0 \chi_1 \widehat{\phi}_0 \ln k \\ &\quad - k^{1/2} \widetilde{V}_0 \chi_1 \mathcal{A}_4(k) (\mathcal{M}_5(g_1) \widehat{g} + g_1) + F. \end{aligned} \quad (5.23)$$

The functional $\mathcal{M}_5(g_1)$ must be chosen using the condition that the right-hand side of the obtained equation is orthogonal to the function $\chi_2 \psi_0$ in $L_2(\mathbb{R}^2)$, which gives

$$\mathcal{M}_5(g_1) = \frac{(F - k^{1/2} \widetilde{V}_0 \chi_1 \mathcal{A}_4(k) g_1, \chi_2 \psi_0)_{L_2(\mathbb{R}^2)}}{\mathcal{M}_2(\widehat{g}) (\widetilde{V}_0 \chi_1 \widehat{\phi}_0, \chi_2 \psi_0)_{L_2(\mathbb{R}^2)} \ln k + k^{1/2} (\widetilde{V}_0 \chi_1 \mathcal{A}_4(k) \widehat{g}, \chi_2 \psi_0)_{L_2(\mathbb{R}^2)}}. \quad (5.24)$$

This functional is well defined, since $\mathcal{M}_2(\widehat{g}) \neq 0$, and integrating by parts we obtain

$$(\widetilde{V}_0 \chi_1 \mathcal{A}_4(k) \widehat{g}, \chi_2 \psi_0)_{L_2(\mathbb{R}^2)} = \int_{\mathbb{R}^2} \overline{\psi}_0 (\mathcal{H} + V_0 - \beta_+) \widehat{\phi}_0 dx' = 2\pi \mathcal{M}_2(\widehat{g}) S_0 \neq 0,$$

where S_0 is defined in (5.9) and is calculated in (5.10). Substituting (5.24) into (5.23) gives rise to an equation for g_1 on the spaces of functions in $L_2(\mathbb{R}^2)$ orthogonal to $\widehat{g}$ and $\chi_2 \psi_0$. The solvability of this equation and its dependence of the solution on k is then easily analyzed on the basis of the Fredholm alternatives. Together with (5.1) this then implies that the resolvent $(\mathcal{H}_0 + V_0 - \beta_+ + k)^{-1}$ is uniformly bounded, as required. Lemma 1 is proved.

Proof of Lemma 2. By (4.34), (4.30), (4.13) and the fact that $\mathcal{L}_5^\varepsilon(k)$ is holomorphic for $k \in \overline{K}_\delta$, the quantity $\mathcal{M}_1(\mathcal{L}_5^\varepsilon(k_\varepsilon) \psi_0)$ can be represented as the sum

$$\begin{aligned} \mathcal{M}_1(\mathcal{L}_5^\varepsilon(k_\varepsilon) \psi_0) &= -\frac{d_+}{4\pi^2} (S_1^\varepsilon - S_2^\varepsilon) + O(\varepsilon^{-1/(\varepsilon\mu_1)}), \\ S_1^\varepsilon &:= \int_{\mathbb{R}^2} \overline{\psi}_0 (V_4 - V_0 - q_1(\varepsilon) V_3 - q_2(\varepsilon) \mathcal{L}_2) \psi_0 dx', \quad S_2^\varepsilon := \int_{\mathbb{R}^2} \overline{\psi}_0 \mathcal{P}_0 \mathcal{L}_1^\varepsilon v^\varepsilon dx', \\ v^\varepsilon &:= \mathcal{B}_1^\varepsilon(k_\varepsilon) h^\varepsilon, \quad h^\varepsilon := (I + \mathcal{P}^\perp \mathcal{L}_1^\varepsilon \mathcal{B}_1^\varepsilon(k))^{-1} \mathcal{L}_3 \psi_0. \end{aligned} \quad (5.25)$$

It follows immediately from (4.14), (4.28) and (4.32) that

$$S_1^\varepsilon = \varepsilon \mu_1 + \varepsilon^2 \int_{\mathbb{R}^2} \left(V_2 + \frac{V_3}{3} \right) |\psi_0|^2 dx' + O(\varepsilon^3). \quad (5.26)$$

We now estimate S_2^ε . The definition of the operator $\mathcal{P}_0$ implies that

$$S_2^\varepsilon = \varkappa_0 \int_{\Omega_\varepsilon} \overline{\psi}_0 \phi_0 \mathcal{L}_1^\varepsilon v^\varepsilon dx'.$$

Since

$$\begin{aligned} \mathcal{L}_1^\varepsilon v^\varepsilon &= e^{i\alpha_1 x_3} \left(\mathcal{H} - \frac{\partial^2}{\partial x_3^2} - \alpha_0^2 - \beta_+ \right) e^{-i\alpha_1 x_3} v^\varepsilon - \left(\mathcal{H} - \frac{\partial^2}{\partial x_3^2} - \alpha_0^2 - \beta_+ \right) v^\varepsilon \\ &= e^{i\alpha_1 x_3} \left(\mathcal{H} - \frac{\partial^2}{\partial x_3^2} + V_0 - \alpha_0^2 - \beta_+ \right) e^{-i\alpha_1 x_3} v^\varepsilon - \mathcal{P}^\perp h^\varepsilon - V_0 v^\varepsilon \end{aligned}$$

and $v^\varepsilon = \mathcal{P}^\perp v^\varepsilon$, $V_0 = V_0(x')$, we obtain

$$S_2^\varepsilon = \kappa_0 \int_{\Omega_\varepsilon} \overline{\psi_0} \theta_0 e^{i\alpha_1 x_3} \left(\mathcal{H} - \frac{\partial^2}{\partial x_3^2} + V_0 - \alpha_0^2 - \beta_+ \right) e^{-i\alpha_1 x_3} v^\varepsilon dx'.$$

We integrate by parts, use the equation for ψ_0 , and again use the fact that $v^\varepsilon = \mathcal{P}^\perp v^\varepsilon$:

$$\begin{aligned} S_2^\varepsilon &= \kappa_0 \int_{\Omega_\varepsilon} e^{-i\alpha_1 x_3} v^\varepsilon e^{i\alpha_1 x_3} \left(\mathcal{H} - \frac{\partial^2}{\partial x_3^2} + V_0 - \alpha_0^2 - \beta_+ \right) e^{i\alpha_1 x_3} \overline{\psi_0} \theta_0 dx' \\ &= -i\kappa_0 \int_{\Omega_\varepsilon} x_3 v^\varepsilon \theta_0 \sum_{i,j=1}^2 \left(\frac{\partial}{\partial x_i} A_{ij} \frac{\partial \alpha_1}{\partial x_j} \overline{\psi_0} + \frac{\partial \alpha_1}{\partial x_i} A_{ij} \left(\frac{\partial \phi^\varepsilon}{\partial x_j} + ix_3 \frac{\partial \alpha_1}{\partial x_j} \phi^\varepsilon \right) \psi_0 \right) dx'. \end{aligned}$$

Since by (4.28), (4.29) and (4.5) we have

$$\|v^\varepsilon\|_{L_2(\Omega_\varepsilon)} \leq C\varepsilon^2 \|h^\varepsilon\|_{L_2(\Omega_\varepsilon)} \leq C\varepsilon^2 \|\mathcal{L}_3 \psi_0\|_{L_2(\Omega_\varepsilon)} \leq C\varepsilon^{5/2},$$

where the constants C are independent of ε , we finally obtain

$$S_2^\varepsilon = O(\varepsilon^3). \quad (5.27)$$

We study now the behaviour of the second summand on the right-hand side of (4.27). As $\mathcal{L}_5^\varepsilon$ is holomorphic in k for $k \in \overline{K}_\delta$, using the asymptotics (4.34) and the definition of $\mathcal{M}_1$ and $\mathcal{A}_2$ we have

$$\begin{aligned} \mathcal{M}_1(\mathcal{L}_5^\varepsilon(k_\varepsilon) \mathcal{A}_2 \mathcal{L}_5^\varepsilon(k_\varepsilon) \psi_0) &= S_3^\varepsilon + O(\varepsilon^{-1/(\varepsilon\mu_1)}), \\ S_3^\varepsilon &:= -\frac{d_+}{4\pi^2} \int_{\mathbb{R}^2} \overline{\psi_0} \mathcal{L}_4^\varepsilon(0) \mathcal{A}_2 \mathcal{L}_5^\varepsilon(0) \psi_0 dx'. \end{aligned}$$

Similarly to the way we obtained (4.30), it is easy to verify that

$$S_3^\varepsilon = -\frac{d_+ \varepsilon^2}{4\pi^2} \int_{\mathbb{R}^2} \overline{\psi_0} V_1 \mathcal{A}_2 V_1 \psi_0 dx' + O(\varepsilon^3).$$

The formula we have obtained and equations (5.25)–(5.27) complete the proof of Lemma 2.

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